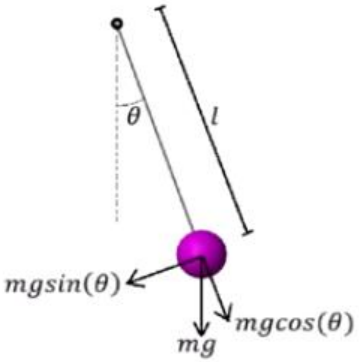
Simple Pendulum

Project Objective

The objective is to:

1. Design a simple pendulum mechanism in **SolidWorks**
2. Derive its mathematical model using physics principles
3. Develop a simulation model in **MATLAB**
4. Analyze:
    - Angular displacement
    - Velocity
    - Acceleration
    - Natural frequency
    - Period of oscillation
5. Compare theoretical and simulation results

Mathematical Modelling



Lagrange Method

Instead of analysing forces, only need:

1. Kinetic Energy (K)
2. Potential Energy (P)

Then use the **Lagrange equation**:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = Q$$

Where,

- $L = K - P$ (The Lagrangian)
- Q = generalized non-conservative force (e.g., damping)

Write the Kinetic Energy

$$K = \frac{1}{2} m l^2 \dot{\theta}^2$$

Write the Potential Energy

$$P = mgl(1 - cos\theta)$$

Form the Lagrangian

$$L = K - P$$

$$L = \frac{1}{2} m l^2 \dot{\theta}^2 - mgl(1 - cos\theta)$$

Compute the Partial Derivatives

Differentiate with respect to angular velocity:

$$\frac{\partial L}{\partial \dot{\theta}} = m l^2 \dot{\theta}$$

Take the time derivative:

$$\frac{d}{dt}(m l^2 \dot{\theta}) = m l^2 \ddot{\theta}$$

Differentiate with respect to angle:

$$\frac{\partial L}{\partial \theta} = -mgl sin\theta$$

Add Damping

The generalized damping force is

$$Q = -b\dot{\theta}$$

Substitute into Lagrange's Equation

$$ml^2\ddot{\theta} - (-mglsin\theta) = -b\dot{\theta}$$

Rearrange:

$$ml^2\ddot{\theta} + b\dot{\theta} + mglsin\theta = 0$$

State Space Model

Define states:

$$x_1 = \theta$$
$$x_2 = \dot{\theta}$$

Therefore:

$$\dot{x_1} = x_2$$
$$\dot{x_2} = -\frac{g}{L}x_1$$

Matrix form:

$$\begin{bmatrix}\dot{x_1}\\ \dot{x_2}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{g}{L} & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$
